

CAPRM-Flood: A Reproducible C++/Python Framework for Property-Level Flood Exposure Indexing

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I. INTRODUCTION

Flooding is the most common weather-related natural disaster [1]. In the northeastern United States, climate change is increasing exposure to flooding, partly through increases in heavy precipitation [2]. Flooding can damage buildings, homes, roadways and water and electrical infrastructure [2]. FEMA estimates that just one inch of flooding can cost up to approximately \$25,000 in damage to a home [3]. Monroe County has been included in 4 FEMA flood-related disaster or emergency declarations from 1956–2022, the most recent of which took place in 2017; further, the Monroe County hazard plan identifies multiple local mechanisms increasing flood susceptibility, including lakeshore flooding and erosion along Lake Ontario, riverine flooding, urban drainage flooding, and flash flooding. FEMA reports that each dollar spent on mitigation saves two dollars in post-flood recovery costs [4].

Property-level flood analysis uses geospatial data detailing mapped flood hazards, terrain, hydrography, precipitation and property location to assess vulnerability and predict loss. FEMA maintains the National Flood Hazard Layer as a database containing geospatial data for flood hazard prediction, including both flood classification and level of risk [5]. The National Flood Insurance Program (NFIP) uses the FEMA Flood Map Service Center (MSC) as its official source for flood hazard information [6]. Beyond FEMA’s MSC, the New York State GIS Program Office provides parcel-centroid data for all 62 New York counties; centroids are generated from the state’s parcel-map program and are mapped to assessment-roll data [7]. When using multiple data sources like these simultaneously, construction of a property-level feature table requires the reconciliation of data sources comprising different schemas, geometry types, coordinate reference systems, and file formats. A spatial join combines observations by geometric relationships instead of a shared identifier or key [8].

Across modern spatial analytics systems driven by spatial joins, research has observed significant variations in performance, demonstrating the need for explicit benchmarking over assumptions about equivalent performance [9]. Established libraries facilitate geospatial feature extraction: GeoPandas

allows for planar spatial operations driven by predicates, such as ‘intersects’ [10]. Because GeoPandas performs planar operations, its geometric and distance calculations depend on appropriate coordinate reference system selection. Shapely offers distinct topological predicates: ‘contains’ excludes points that lie only on a polygon boundary, whereas ‘covers’ includes geometries lying within either the interior or boundary of the selected geometry [11]. A common structure across indexed spatial processing systems is the filter-and-refine structure, which first searches bounding boxes and then applies a predicate to filter candidate geometries [12]. Boundary-point results may differ under different predicate semantics or across different spatial analytics libraries; in general, large batches benefit from spatial indexes that reduce the number of exact comparisons needed to identify matching geometry. These traits suggest the need for separate evaluations of performance and correctness in a spatial feature-extraction system. This project seeks to answer the following question: Can a reproducible C++/Python pipeline extract property-level flood-hazard features from public geospatial data while preserving agreement with a trusted Python baseline and providing competitive large-batch spatial-processing performance?

This paper presents CAPRM-Flood, a benchmarked spatial feature-extraction pipeline for property-to-hazard joins. The study region is Monroe County and Rochester, New York. The system takes as input a stable property identifier and coordinates and outputs traceable per-property evidence fields, normalized components, and a relative exposure index. The system comprises two layers of processing driven respectively by Python and C++. Python responsibilities include ingestion, schema cleaning, normalization, GeoPandas/Shapely reference computation, validation and reporting; C++ responsibilities include indexed batch point-in-polygon processing and relevant flood-risk feature lookups, such as distance to the nearest mapped water source.

CAPRM-Flood presents a reproducible workflow for deriving traceable property-level features from public property and flood-hazard data. It provides a C++ spatial-processing core for batch property-to-hazard lookups and a validation framework for comparing its correctness and performance

against GeoPandas and Shapely, including point-in-polygon agreement and boundary-case analysis. Finally, it provides a performance evaluation framework with distinct benchmarks for ingestion, index construction, query time, throughput, memory use and scaling. These components provide transparent screening and feature infrastructure for reproducible research and the preparation of validated data for later risk analysis or modeling.

II. RELATED WORK

Works related to CAPRM-Flood can be grouped according to function. As FEMA's database of current effective flood-hazard data, the NFHL provides flood-hazard geometries and regulatory mapping information derived from effective flood maps and Letters of Map Change (LOMC) [5]. FEMA's National Risk Index offers baseline risk measurements for flooding among 17 other natural hazards at the community level, and is accessible to the public through the Resilience Analysis and Planning Tool (RAPT) [13]. Additionally, FEMA provides Hazus as a standardized set of hazard-estimation tools and corresponding data; Monroe County's 2023 Hazard Mitigation Plan (HMP) utilizes Hazus to predict loss and damages due to flooding [4], [14]. Hazus flood-model methodology is implemented in FAST to swiftly analyze building-level flood risk [15]. These resources supply hazard information or perform broader risk analysis, whereas CAPRM-Flood investigates reproducible computations of property-level features with transparency in both feature derivation and performance evaluation.

Multi-Criteria Decision Analysis (MCDA) provides an established set of methodologies for assessing flood risk. In particular, the Analytic Hierarchy Process (AHP) facilitates the mapping of relative weights to different geospatial criteria [16]. GIS-based multi-criteria analysis has been used to map flood vulnerability in the trans-boundary Shatt Al Arab basin (Iraq-Iran) [17]. Factors used in MCDA studies include slope, elevation, land use, river proximity, drainage density, rainfall and soil type, while AHP facilitates a structured comparison of each criterion's influence on flood vulnerability [18]. CAPRM-Flood adopts configurable weights and explicit sensitivity analysis to evaluate the stability of property rankings under different weighting assumptions, while separately evaluating implementation correctness and computational performance.

CAPRM-Flood evaluates correctness using GeoPandas and Shapely as a Python reference environment. GeoPandas provides predicate-driven spatial joins and relies on Shapely for performing geometric operations [10]. Existing tools facilitate spatial feature extraction in C++, though prior systems research demonstrates significant performance variation across spatial systems and workloads [9]. Boost.Geometry's R-tree implementation and spatial query interface are feasible components of a C++ core for executing indexed spatial queries [19]. In leveraging these resources, CAPRM-Flood contributes a bounded case study involving property points and real FEMA polygon geometries.

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